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(54) **FISHING HOOK REMOVER AND VENT
TOOL**

(52) **U.S. Cl.**
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(57) **ABSTRACT**

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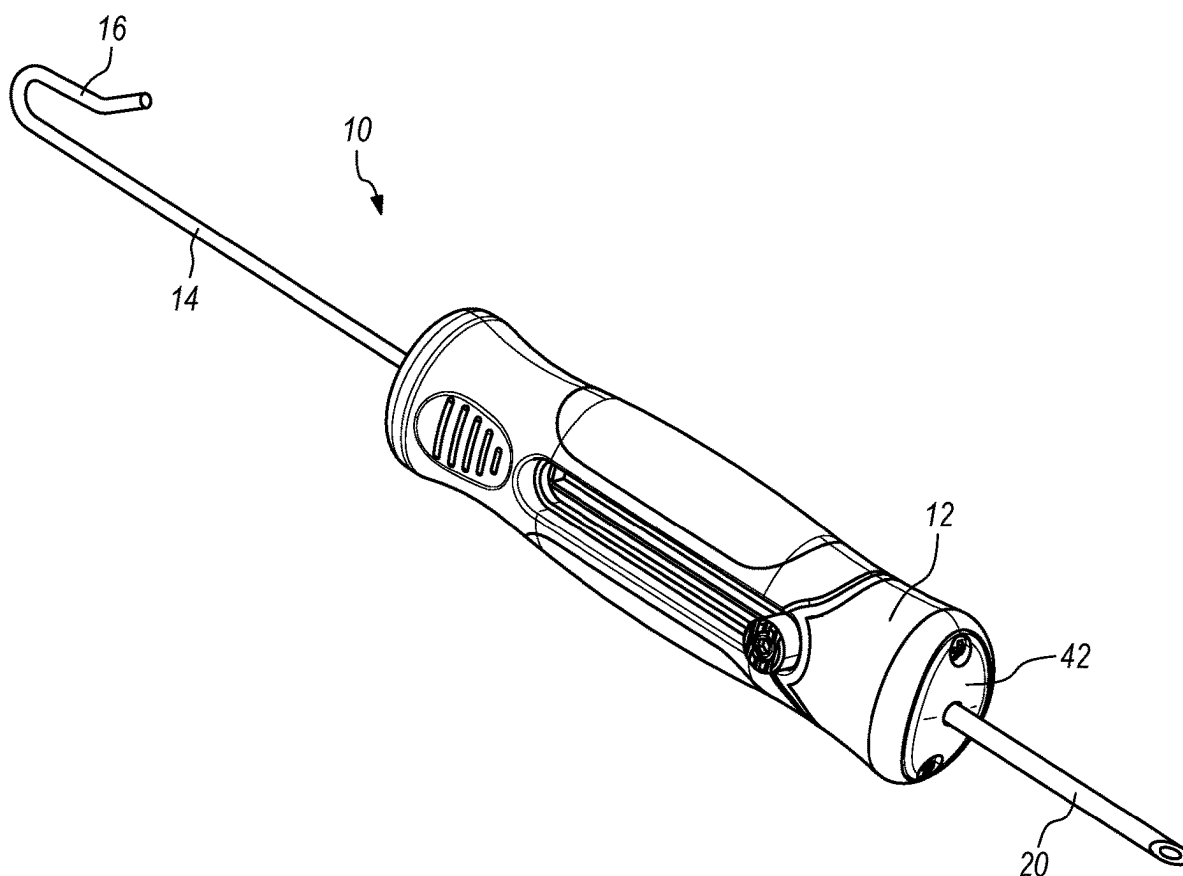
A fishing hook remover and vent tool is disclosed. The tool includes a handle having a first end and a second end, a hook remover connected to the first end of said handle, and a vent needle assembly. The vent needle assembly includes a handle insert positioned in an internal compartment of the handle and a slidable needle assembly positioned in the interior of the handle insert. The slidable needle assembly includes a vent needle, which is slidable in and out of the second end of the handle.

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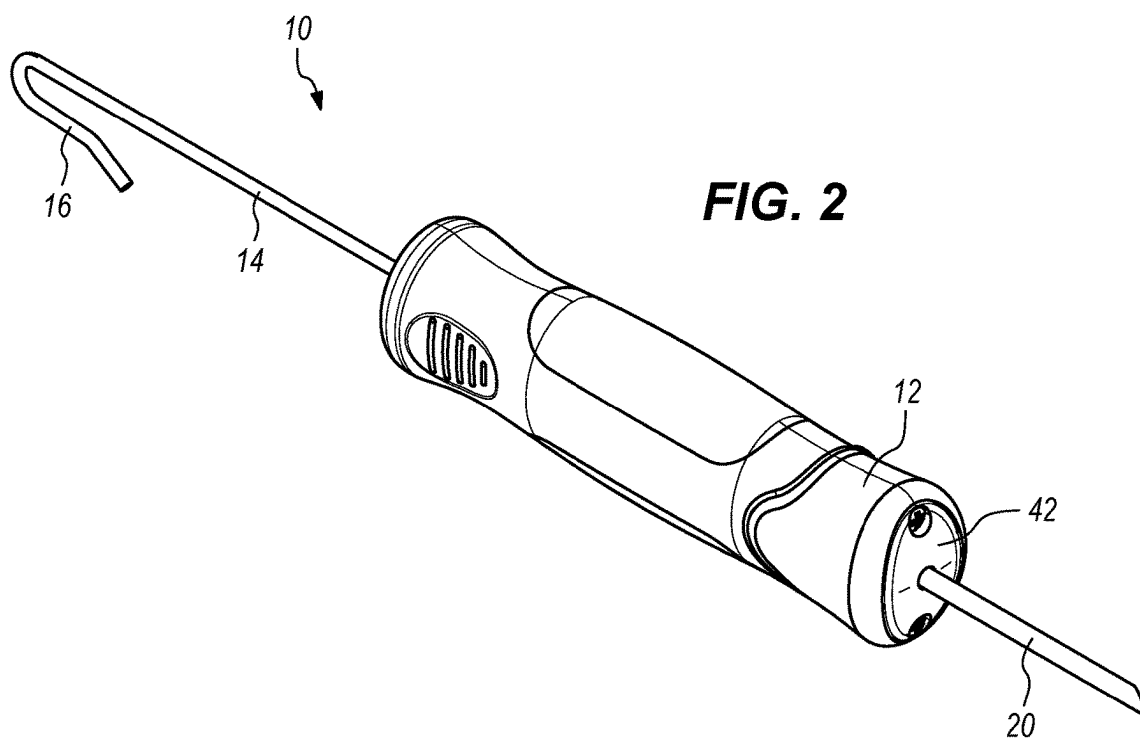
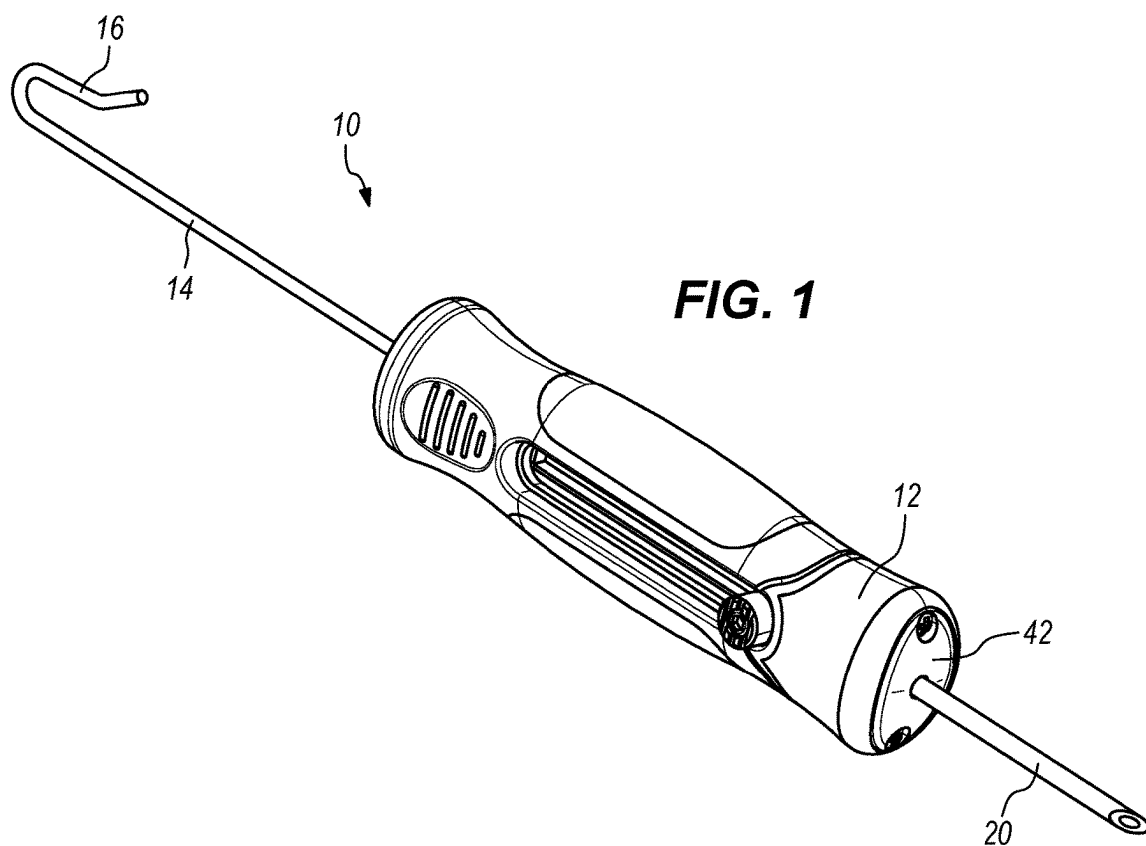


FIG. 3

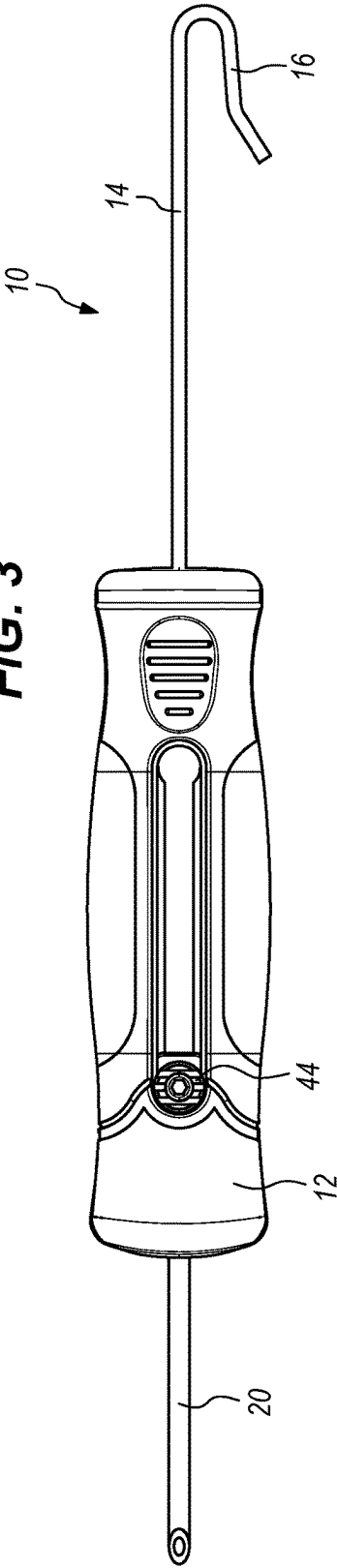


FIG. 4

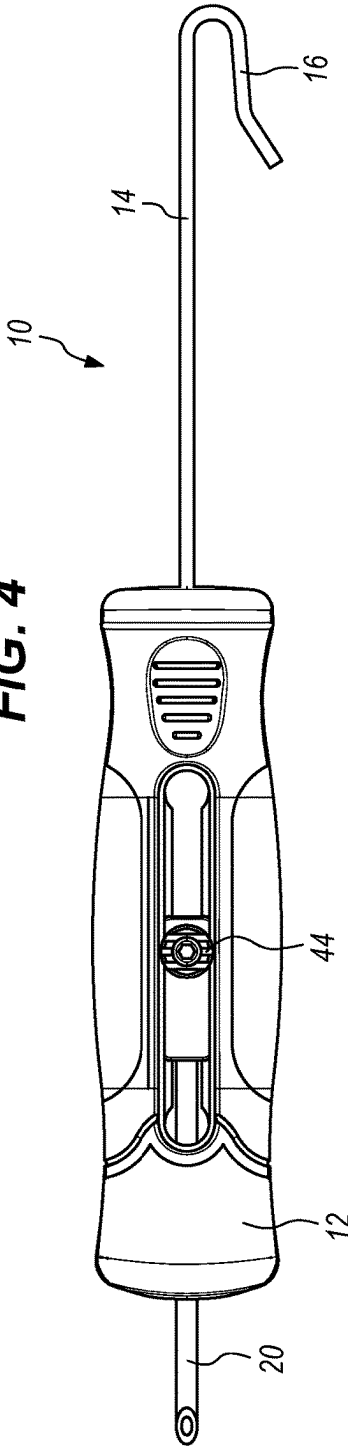
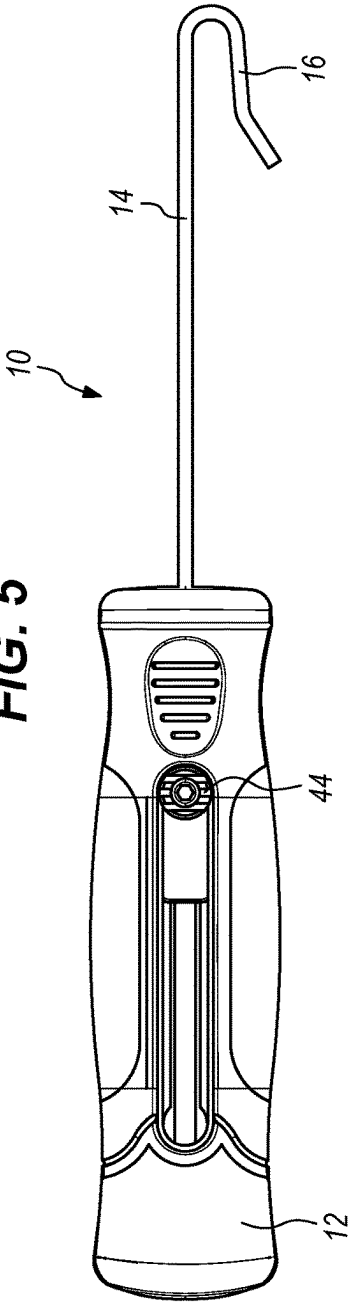


FIG. 5



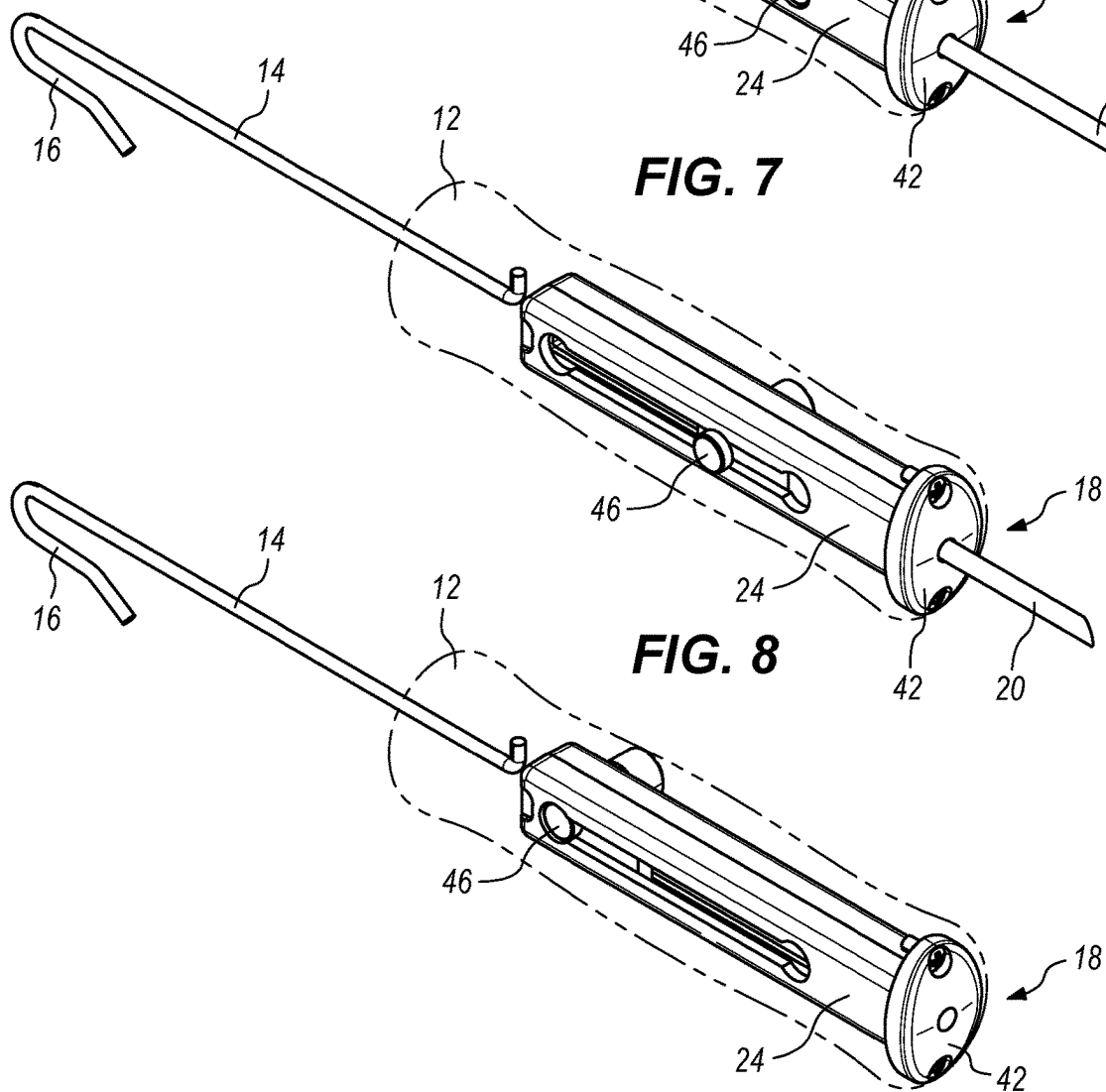
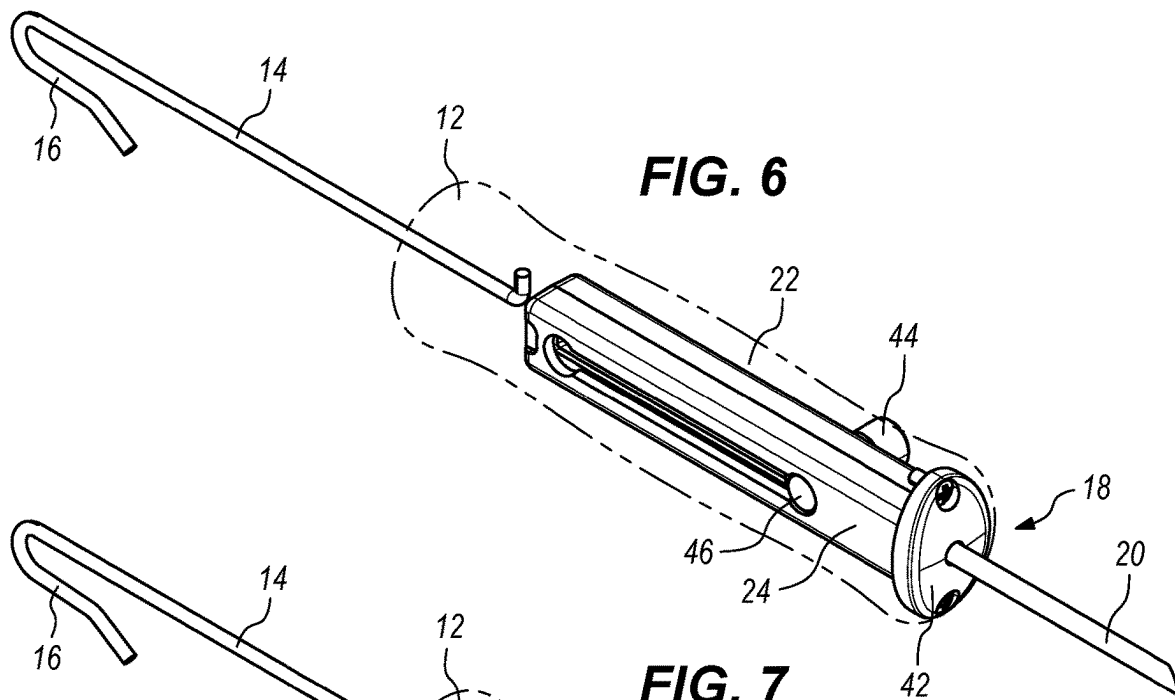


FIG. 9

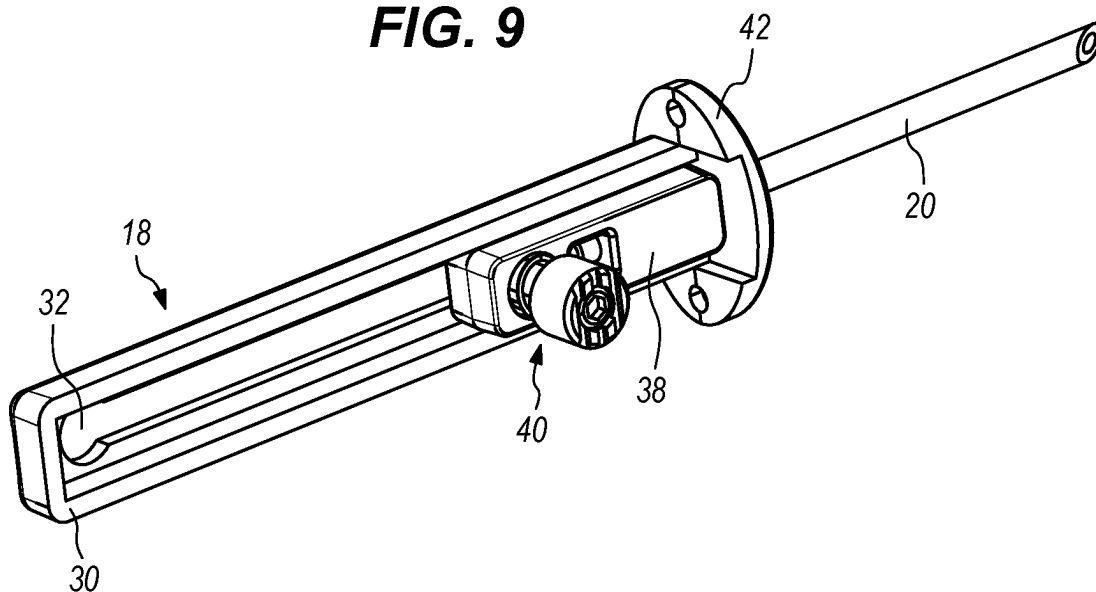


FIG. 10

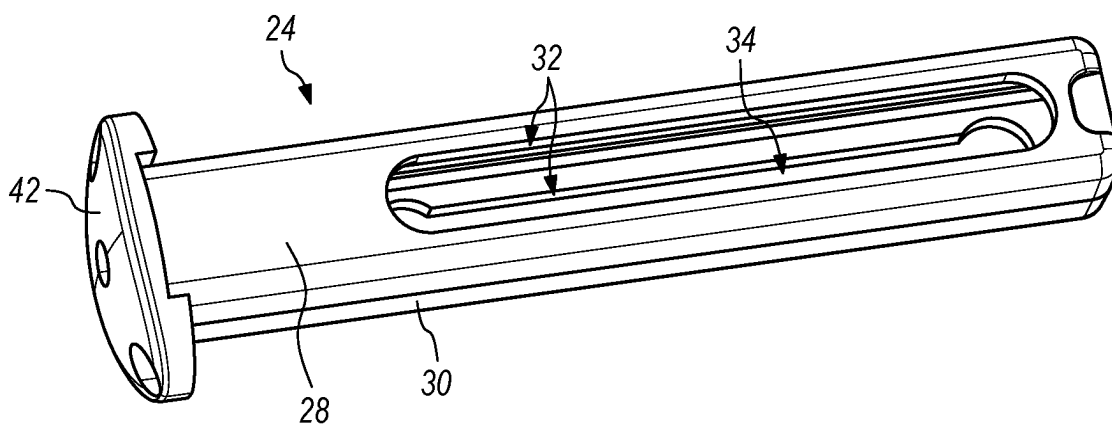


FIG. 11

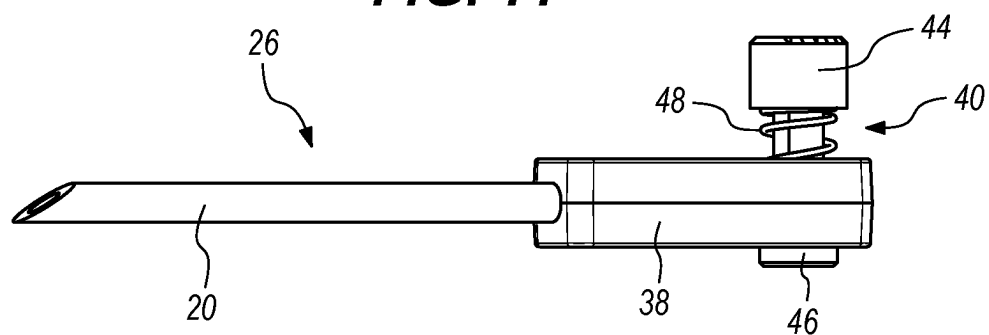


FIG. 12

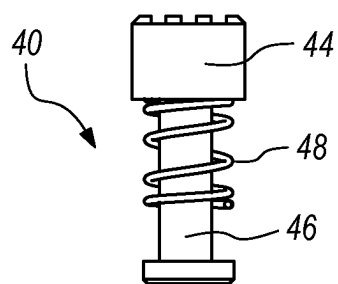


FIG. 13

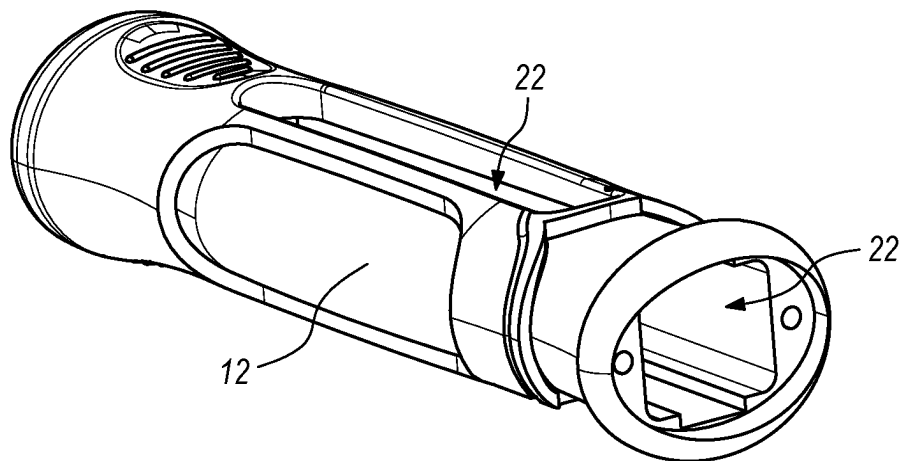


FIG. 14

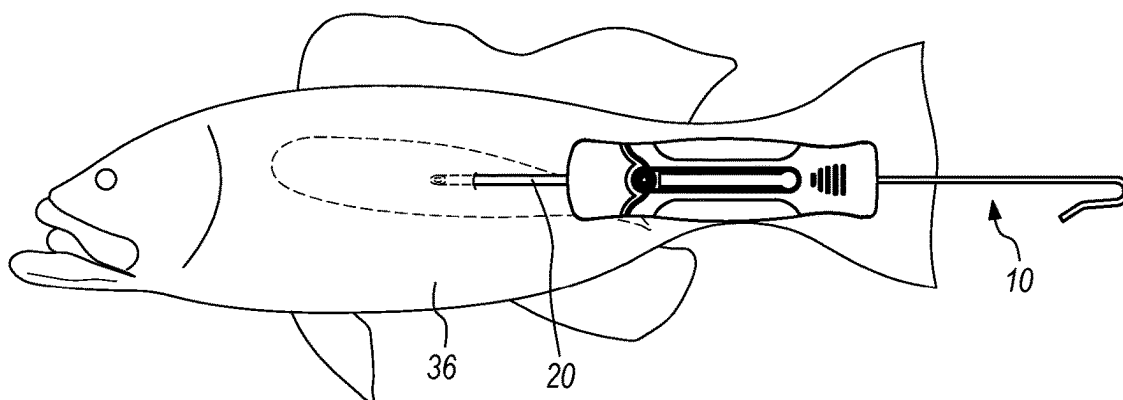
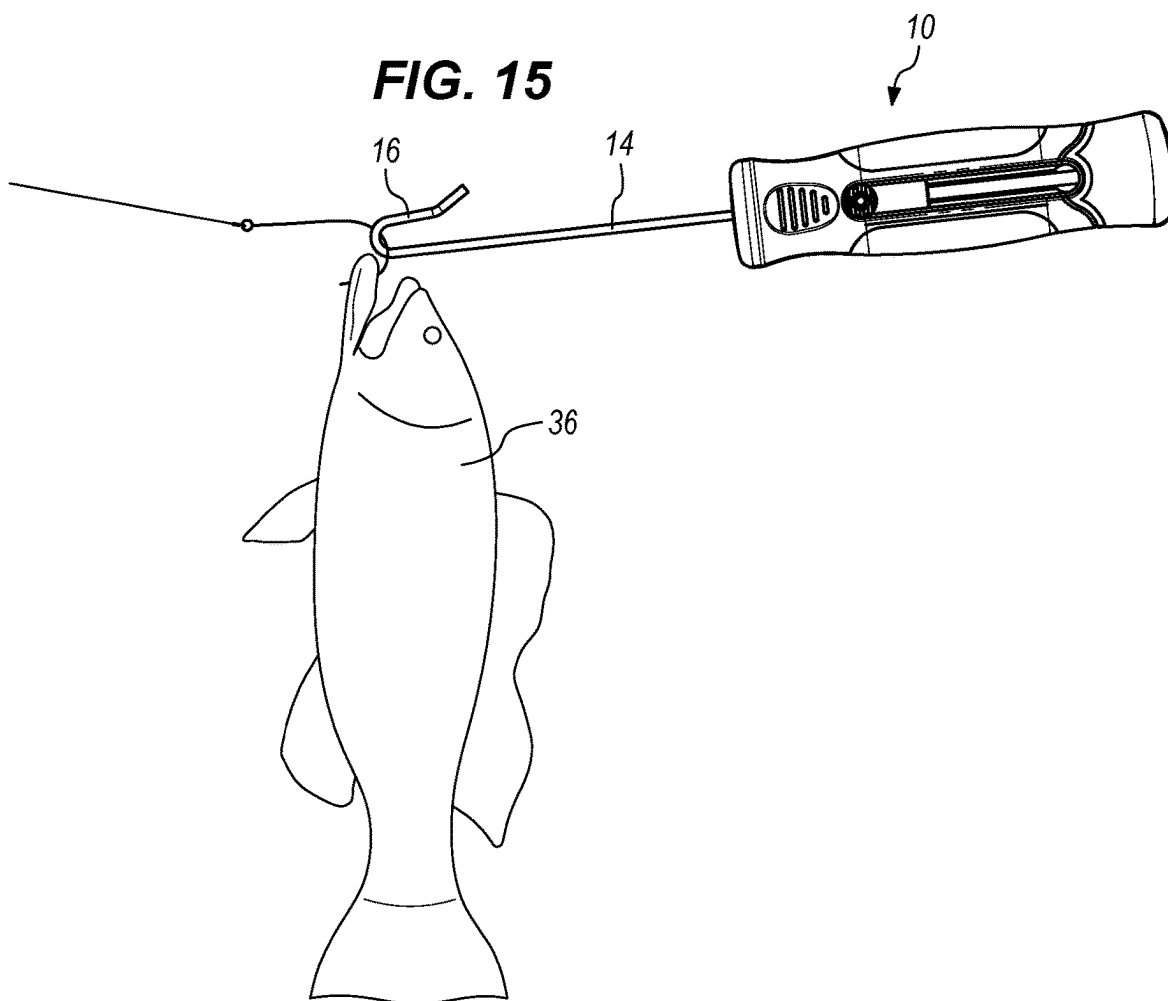


FIG. 15



FISHING HOOK REMOVER AND VENT TOOL

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not applicable

BACKGROUND OF THE INVENTION

[0003] Fishermen must be prepared for many situations that they may encounter while fishing, so it is common for fishing tackle boxes to be packed full of fishing supplies and gadgets. But tackle box space is at a premium, so small tools capable of performing different tasks are preferred. One challenge faced by nearly all of fishermen is safely removing the fishing hook from the mouth of the fish. In some instances, the fisherman may desire to remove the hook without touching the fish to prevent injury to himself or the fish. To overcome this challenge, some fishermen keep a dehooking or hook remover device in their tackle box. Another challenge faced by some fishermen is the need to release or vent gases out of the swim bladder of fish caught at depth before the fish are released. In these situations, if the gases in the swim bladder are not released, then the fish will be prevented from returning quickly to depth and may die as a result. Inserting a hypodermic needle into the swim bladder is a common method for releasing the gases, but the needle can easily be damaged in a tackle box and the fishermen risks pricking himself with the sharp needle left in the tackle box.

[0004] It would therefore be desirable to develop a tool that can be used to both remove the fishing hook from the mouth of the fish and to vent the fish. It would also be desirable for the vent needle to be stowed in the tool when it is not in use to prevent damage to the needle and injury to the fisherman.

BRIEF SUMMARY OF THE INVENTION

[0005] The present invention is directed to a fishing hook remover and vent tool. The tool includes a handle having a first end and a second end, a fishing hook remover connected to the first end of the handle, and a fishing vent needle assembly. The vent needle assembly includes a handle insert positioned in an internal compartment of the handle and a slidable needle assembly positioned in the interior of the handle insert. The slidable needle assembly comprises a vent needle, which is slidable in and out of the second end of the handle.

[0006] These and other features, objects and advantages of the present invention will become better understood from a consideration of the following detailed description of the preferred embodiments and appended claims in conjunction with the drawings as described following:

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a top perspective view of the fishing hook remover and vent tool of the present invention.

[0008] FIG. 2 is a bottom perspective view of the fishing hook remover and vent tool of the present invention.

[0009] FIG. 3 is a top view of the fishing hook remover and vent tool of the present invention with the vent needle in the open position.

[0010] FIG. 4 is a top view of the fishing hook remover and vent tool of the present invention with the vent needle transitioning between the open position and the closed position.

[0011] FIG. 5 is a top view of the fishing hook remover and vent tool of the present invention with the vent needle in the closed position.

[0012] FIG. 6 is a bottom perspective view of the fishing hook remover and vent tool of the present invention with the vent needle in the open position and the handle in outline.

[0013] FIG. 7 is a bottom perspective view of the fishing hook remover and vent tool of the present invention with the vent needle transitioning between the open position and the closed position and the handle in outline.

[0014] FIG. 8 is a bottom perspective view of the fishing hook remover and vent tool of the present invention with the vent needle in the closed position and the handle in outline.

[0015] FIG. 9 is a top perspective view of the vent needle assembly of the fishing hook remover and vent tool of the present invention with the top plate omitted.

[0016] FIG. 10 is a top perspective view of the handle insert of the fishing hook remover and vent tool of the present invention with the slidable needle assembly omitted.

[0017] FIG. 11 is a side view of the slidable needle assembly of the fishing hook remover and vent tool of the present invention.

[0018] FIG. 12 is a side view of the push button of the fishing hook remover and vent tool of the present invention.

[0019] FIG. 13 is a top perspective view of the handle of the fishing hook remover and vent tool of the present invention with the handle insert and the slidable needle assembly omitted.

[0020] FIG. 14 is a perspective view of a fish with the swim bladder of the fish being punctured by the vent needle of the fishing hook remover and vent tool of the present invention.

[0021] FIG. 15 is a perspective view of a caught fish with the hook in the caught fish's mouth interlocked with the hook remover of the fishing hook remover and vent tool of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0022] With reference to FIGS. 1-15, the preferred embodiments of the present invention may be described. The fishing hook remover and vent tool 10 of the present invention includes a handle 12 with a hook remover 14 connected to one end of the handle 12 and a vent needle 20 slidably connected to the other opposite end of the handle 12, as shown in FIGS. 1-2.

[0023] The handle 12 of the tool 10 is used for grasping the tool. The handle 12 preferably includes an open internal compartment 22. One end of the handle includes an opening that receives one end of the hook remover 14. In one embodiment, the hook remover 14 is preferably an austenitic or martensitic steel rod, but it should be understood that other strong and durable materials may alternatively be used. One end of the hook remover 14 is bent at approximately a 90 degree angle to prevent the hook remover 14 from being removed from the opening in the handle 12, as shown in FIGS. 6-8. At the opposite end, the hook remover

14 includes a hook end **16** at the end opposite of the handle **12**. The hook remover **14** extends away from the end of the handle **12** before forming the hook end **16**. The hook end is formed by the rod curving back towards the handle. The terminal end portion of the rod is angled at approximately 45-degrees from the adjacent portion of the rod that extends back towards the end of the handle **12**.

[0024] In use, while holding the fishing line attached to the caught fish **36** with one hand, the fisherman uses his other hand to interlock the fishing hook in the fish's mouth with the hook end **16** of the hook remover of the tool **10**, as shown in FIG. 15. By rotating the handle **12** of the tool **10** and thus rotating the hook end **16** of the hook remover **14**, the fisherman is able to rotate the fishing hook in the fish's mouth. The handle **12** of the tool **10** is rotated until the fishing hook in the fish's mouth is facing downward. As a result, the fish falls off of the fishing hook. In some circumstances, the fisherman may softly jiggle the handle **10** to aid the fish sliding off the hook.

[0025] The tool **10** includes a vent needle assembly **18**. The vent needle assembly **18** of the tool includes the vent needle **20** that is used by fishermen to release or vent gases out of the swim bladder of fish caught at depth before they are released, as shown in FIG. 14. The vent needle **20** is slidable into the internal compartment **22** of the handle **12** for stowing and slidable out of the handle **12** when the vent needle **20** is being used. Stowing the needle **20** prevents damage to the needle and potential injury to the fisherman.

[0026] The vent needle assembly **18** includes a handle insert **24** and a slidable needle assembly **26**. The slidable needle assembly **26** includes the needle **20**, a connector **38**, and a push button **40**. The needle **20** has a pointed end and a non-pointed end. The non-pointed end is received by the connector **38**. In the preferred embodiment, the non-pointed end is received through an opening in the side of the connector **38**, as shown in FIG. 11. The pointed end of the needle **20** therefore extends away from the connector **38**. The push button **40** includes a thumb press **44**, a rod **46**, and a spring **48**. The spring **48** surrounds the rod **46**, and the rod **46** passes through a vertically-oriented opening in the connector **38**, as shown in FIG. 11. The thumb press **44** fits on top of the rod **46** that is positioned outside of the top of the connector **38**. The bottom end of the rod **46** has a larger diameter than the opening in the connector **38** to prevent the rod from being removed from the top of the connector **38**.

[0027] The handle insert **24**, which is positioned in the open internal compartment **22** of the handle **12**, includes a top plate **28** and a bottom plate **30**. The top plate **28** and the bottom plate **30** are connected along their perimeter. Each of the top plate **28** and the bottom plate **30** includes a slot that extends a portion of the length of the plate. The slot of the top plate **28** is a long oval shape, as shown in FIGS. 3-5 and 10. The slot of the bottom plate **30** is elongated with the ends being rounded, as shown in FIGS. 6-8 and 10. The slots **32** of the top plate **28** and the bottom plate **30** are aligned to form a track **34**. A portion of the rod of the needle assembly **26** is slidable in the track **34** of the handle insert **24**. The end of the handle insert **24** includes an end plate **42**. The end plate **42** closes the end of the internal compartment **22** in the handle **12**. In one embodiment, the end plate **42** is fastened to the end of the handle **12** with fasteners, such as screws.

[0028] Moving the slidable needle assembly **26** within the track **34** moves the vent needle **20** from the closed or stowed position in the handle **12** to the open or usable position

outside of the handle **12**. When the vent needle **20** is in the closed or stowed position, the push button **40** is at the end of the track **34** nearest the hook remover **14**, as shown in FIG. 5. In this position, as shown in FIG. 8, the bottom end of the rod **46** with the larger diameter is positioned in the rounded end of the bottom plate **30** of handle insert **24**. Because the bottom portion of the rod **46** is too large to move within the elongated portion of the slot of the bottom plate **30**, the rod **46** (and the other components of the slidable needle assembly **26**) are locked in place in the stowed position.

[0029] When the vent needle **20** is needed, the fisherman pushes down on the thumb press **44**, which causes the spring to compress and the rod **46** to move downward. In this regard, the rod **46** extends further downward through the bottom of the connector **38**. The rod **46** moves downward until the larger-diameter bottom end of the rod **46** is entirely below the rounded end of the bottom plate **30**, as shown in FIG. 7. Because the bottom end of the rod **46** is no longer restricted from movement by the bottom plate **30**, the fisherman can now slide the rod **46** along the length of the track **34** by sliding the thumb press **44** until the rod reaches the end of the track **34**, which is the end of the slots **32**. As the rod is being slid towards the end of the handle, the vent needle **20** that is connected to the connector **38** is pushed through an opening in the end plate **42**. Once the rod reaches the end of the track **34**, the fisherman may remove the downward pressure on the thumb press **44**. This causes the spring **48** to decompress and the rod **46** to move upward. As a result, the larger-diameter bottom end of the rod **46** pops into the rounded end of the bottom plate **30**, as shown in FIG. 6. Because the bottom end of the rod **46** is too large to move within the elongated portion of the slot of the bottom plate **30**, the rod **46** (and the other components of the slidable needle assembly **26**) are locked in place in the open position.

[0030] The vent needle **20** may also be moved back to the closed or stowed position in the handle **12** when the needle is no longer being used. As shown in FIG. 6, when the vent needle **20** is in the open position, the push button **40** is at the end of the track **34** farthest from the hook remover **14**. In this position, the bottom end of the rod **46** with the larger diameter is positioned in the rounded end of the bottom plate **30** of handle insert **24**. Because the bottom end of the rod **46** is too large to move within the elongated portion of the slot of the bottom plate **30**, the rod **46** (and the other components of the slidable needle assembly **26**) are locked in place in the open position. When the vent needle **20** is no longer needed, the fisherman pushes down on the thumb press **44**, which causes the spring to compress and the rod **46** to move downward. In this regard, the rod **46** extends further downward through the bottom of the connector **38**. The rod **46** moves downward until the larger-diameter bottom end of the rod **46** is entirely below the rounded end of the bottom plate **30**. Because the bottom portion of the rod **46** is no longer restricted from movement by the bottom plate **30** as shown in FIG. 7, the fisherman can now slide the rod **46** along the length of the track **34** by sliding the thumb press **44** until the rod reaches the end of the track **34**, which is the end of the slots **32**. As the rod is being slid towards the hook remover, the length of the vent needle **20** that is exposed through opening in the end plate **42** is reduced. Once the rod reaches the end of the track **34**, the fisherman may remove the downward pressure on the thumb press **44**. This causes the spring **48** to decompress and the rod **46** to move upward. As

a result, the larger-diameter bottom end of the rod **46** pops into the rounded end of the bottom plate **30**, as shown in FIG. **8**. Because the bottom end of the rod **46** is too large to move within the elongated portion of the slot of the bottom plate **30**, the rod **46** (and the other components of the slidable needle assembly **26**) are locked in place in the stowed or closed position.

[0031] In use, the vent needle **20** of the vent needle assembly **18** is used by the fisherman to release or vent gases out of the swim bladder of a caught fish **36**. As shown in FIG. **14**, with the vent needle **20** in the open position, the fisherman punctures the swim bladder of the fish **36**, which allows accumulated gases to be released from the swim bladder. After the swim bladder is punctured by the needle **20**, the fisherman may place the fish back into the water to swim freely.

[0032] The present invention has been described with reference to certain preferred and alternative embodiments that are intended to be exemplary only and not limiting to the full scope of the present invention as set forth in the appended claims.

We claim:

1. A fishing hook remover and vent tool comprising:
a handle having a first end and a second end;
a hook remover connected to said first end of said handle;
and
a vent needle, wherein said vent needle is slidable in and out of said second end of said handle.
2. The fishing hook remover and vent tool of claim **1**, further comprising a vent needle assembly, wherein said vent needle assembly comprises a handle insert positioned in an internal compartment of said handle and a slidable needle assembly positioned in an interior of said handle insert.
3. The fishing hook remover and vent tool of claim **2**, wherein said slidable needle assembly comprises said vent needle.
4. The fishing hook remover and vent tool of claim **3**, wherein said slidable needle assembly further comprises a connector and a push button, wherein said vent needle and said push button are connected to said connector.
5. The fishing hook remover and vent tool of claim **2**, wherein said handle insert comprises a top plate with a top slot and a bottom plate with a bottom slot.
6. The fishing hook remover and vent tool of claim **4**, wherein said slidable needle assembly is operable to manually slide said vent needle in and out of said second end of said handle.

7. The fishing hook remover and vent tool of claim **4**, wherein said push button comprises a thumb press, a rod, and a spring, wherein said spring surrounds said rod and said thumb press is connected to a top of said rod.

8. The fishing hook remover and vent tool of claim **5**, wherein said top slot and said bottom slot are aligned and form a track, wherein said slidable needle assembly is slidable in said track.

9. A fishing hook remover and vent tool comprising:

- a handle having a first end and a second end;
- a hook remover connected to said first end of said handle;
- and
- a vent needle assembly, wherein said vent needle assembly comprises a handle insert positioned in an internal compartment of said handle and a slidable needle assembly positioned in an interior of said handle insert, wherein said slidable needle assembly comprises a vent needle, wherein said vent needle is slidable in and out of said second end of said handle.

10. The fishing hook remover and vent tool of claim **9**, wherein said slidable needle assembly further comprises a connector and a push button, wherein said vent needle and said push button are connected to said connector.

11. The fishing hook remover and vent tool of claim **10**, wherein said push button comprises a thumb press, a rod, and a spring, wherein said spring surrounds said rod and said thumb press is connected to a top of said rod.

12. The fishing hook remover and vent tool of claim **11**, wherein said handle insert comprises a top plate with a top slot and a bottom plate with a bottom slot, wherein said bottom slot is elongated with a first rounded end and a second rounded end.

13. The fishing hook remover and vent tool of claim **12**, wherein said top slot and said bottom slot are aligned and form a track, wherein said slidable needle assembly is slidable in said track when a bottom end of said rod extends below said bottom slot.

14. The fishing hook remover and vent tool of claim **10**, wherein said push button is configured to manually slide said vent needle in and out of said second end of said handle.

15. The fishing hook remover and vent tool of claim **10**, wherein said rod is slidable in said connector.

16. The fishing hook remover and vent tool of claim **13**, wherein said vent needle is locked in position when said bottom end of said rod is positioned in said first rounded end or said second rounded end of said bottom slot.

* * * * *